

```
class Student:
```

```
    def __init__(self, name, student_id, email, age, department):
```

```
        self.name = name
```

```
        self.student_id = student_id
```

```
        self.__email = email
```

```
        self.age = age
```

```
        self.department = department
```

```
        self.__marks = []
```

```
    def display_info(self):
```

```
        print("\nStudent Information\n")
```

```
        print("Name:", self.name)
```

```
        print("Student ID:", self.student_id)
```

```
        print("Email:", self.__email)
```

```
        print("Age:", self.age)
```

```
        print("Department:", self.department)
```

```
        print("Student Type:", self.get_student_type())
```

```
    def calculate_result(self, *marks):
```

```
        self.__marks = marks
```

```
        if len(marks) == 0:
```

```
            print("No marks entered.")
```

```
            return
```

```
        total = sum(marks)
```

```
average = total / len(marks)
```

```
print("Total Marks:", total)
```

```
print("Average Marks:", average)
```

```
if average >= 80:
```

```
    print("Grade: A+")
```

```
elif average >= 70:
```

```
    print("Grade: A")
```

```
elif average >= 60:
```

```
    print("Grade: B")
```

```
elif average >= 50:
```

```
    print("Grade: C")
```

```
elif average >= 40:
```

```
    print("Grade: D")
```

```
else:
```

```
    print("Grade: F")
```

```
def get_student_type(self):
```

```
    return "Student"
```

```
def get_email(self):
```

```
    return self.__email
```

```
class UndergraduateStudent(Student):
```

```
def __init__(self, name, student_id, email, age, department, semester):  
    super().__init__(name, student_id, email, age, department)  
    self.semester = semester
```

```
def get_student_type(self):  
    return "Undergraduate Student"
```

```
def display_info(self):  
    super().display_info()  
    print("Semester:", self.semester)
```

```
class GraduateStudent(Student):
```

```
    def __init__(self, name, student_id, email, age, department, research_topic):  
        super().__init__(name, student_id, email, age, department)  
        self.research_topic = research_topic
```

```
    def get_student_type(self):  
        return "Graduate Student"
```

```
    def display_info(self):  
        super().display_info()  
        print("Research Topic:", self.research_topic)
```

```
student1 = UndergraduateStudent(
```

```
"Richi",  
"12345",  
"richa@example.com",  
21,  
"CSE",  
5  
)
```

```
student2 = GraduateStudent(  
    "Sadia",  
    "54321",  
    "sadia@example.com",  
    24,  
    "CSE",  
    "Artificial Intelligence"  
)
```

```
student1.display_info()  
student2.display_info()
```

```
print("\nUndergraduate Result\n")  
student1.calculate_result(85, 78, 90, 88)
```

```
print("\nGraduate Result\n")  
student2.calculate_result(75, 82, 79)
```

```
print("\nPolymorphism\n")
```

```
students = [student1, student2]
```

```
for student in students:
```

```
    print(student.get_student_type())
```